

IONUT MORARU (YANI)

Curriculum Vitae

EDUCATION

2017 – 2022 **PhD in Computer Science Research**

KING'S COLLEGE LONDON, UK

*Supervisors: Prof. Simon Parsons, Prof. Stefan Edelkamp,
Prof Peter McBurney, Dr Andrew Coles*

2013 – 2017 **BSc in Computer Science with a Year Abroad**

KING'S COLLEGE LONDON, UK

Faculty of Natural, Mathematical & Engineering Sciences

YEAR ABROAD: NATIONAL UNIVERSITY OF SINGAPORE

Grade achieved: 2:1 – Supervisor: Dr. Christian Urban

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EMPLOYMENT

2024 – **University of Lincoln - Lecturer in Intelligent Robotics and AI**

Delivered Introduction to Data Mining (AGR9013)
and Robotics and Sustainability (ROB2003)

2023 – 2024 **University of Lincoln - Post Doctoral Research Associate in Intelligent Robotics**

Worked as a Research Associate on delivering several projects,
assisted with the teaching of Data Mining and Research Methods

2024 – 2026 **AGRI-OPENCORE – INNOVATE UK**

Working on the autonomy and usability of the system

2024 – 2026 **HiRES SOILS – INNOVATE UK**

Involved from the proposal process, project developing a robot to sample
and evaluate the soils of crop fields at a higher resolution.

2021 – 2023 **University of Lincoln - Post Doctoral Research Associate in Explainable AI**

Worked as a Research Associate on delivering several projects,
assisted with the teaching of Data Mining and Research Methods

2021 – 2024 **CONSULT PROJECT – EPSRC**

Research project focused on combining wearable health technology with an AI assistant
based on Argumentation for supporting stroke survivors in their recovery.
Developed and deployed new back-end of a the cloud based system supporting the project.

2022 – 2024 **ARWAC ATTACK BLACKGRASS IN FARMING PROJECT– INNOVATE UK**

Project in partnership with ARWAC for designing and developing an autonomous
platform to identify and remove blackgrass from crop fields.

2021 **King's College London - THuMP Research Internship – 2 months**

*Developed the simulator and prepared a user study for evaluating the level of explanations
a robotic agent needs to provide when working with in office environments.*

2016 **Volkswagen Data:Lab - Technology and Data Science Internship – 5 months**

*Introduced ReactJS and React-Native to the Laboratory,
reviewed and added new technologies in the VW tech library and
I also maintained some of the datasets for the Data Science team*

ACADEMIC PUBLICATIONS & RESEARCH INTERESTS

- 2025 **Miscanthus-AI: Applying computational argumentation for decision support in plant breeding** - Hardy, O., Xydis, A., Su, Y., Nazari, K., **Moraru, I.**, Gaju, O., Pearson, S. and Sklar, E.I., in *The Plant Phenome Journal* 8.1 (2025): e70028.
- 2024 **ArgHRI: Mediating Human-Robot Teaming With Automated Argumentation in Agriculture** - Moraru, I., Xydis, A., Huang, Z., & Sklar, E. I.
In *2024 IEEE 2nd Conference on AgriFood Electronics (CAFE)* (pp. 190-194). IEEE.
- 2024 **IRfold: An RNA Secondary Structure Prediction Approach** - AIAI24
âHurst, D., Iliopoulos, C. S., Lim, Z., **Moraru, I.**. In *IFIP International Conference on Artificial Intelligence Applications and Innovations* (pp. 131-144).
- 2023 **A Variable Autonomy approach for an Automated Weeding Platform** - VAT-HRI
Moraru, I., Tsvetan, Z., Coutts, S., Li, D., Sklar, E.I.,
In the Variable Autonomy for Human-Robot Teaming (VAT) workshop at the 18th ACM/IEEE International Conference on Human-Robot Interaction (HRI)
- 2022 **Using IoT Assistive Technologies for Older People Non-Invasive Monitoring and Living Support in Their Homes** - IJERPH
Moraru, S.A., Mosoi, A.A., Kristaly, D.M., **Moraru, I.**, Petre, V.Ş., Ungureanu, D.E., Perniu, L.M., Rosenberg, D. and Cocuz, M.E.
In the no. 10 edition *International Journal of Environmental Research and Public Health*.
- 2021 **Using Plan Libraries for Improved Plan Execution** - UKRAS 2021
Moraru, I., Canal, G. and Parsons, S.
Proc. of the *UKRAS21 Conference : "Robotics at home"*.
- 2019 **Simplifying Automated Pattern Selection for Planning with symbolic pattern databases** - KI 2019
Moraru, I., Edelkamp, S., Franco, S. and Martinez, M.
Proc. of the *Joint German/Austrian Conference on Artificial Intelligence (Künstliche Intelligenz)*.
- 2019 **Benchmarks Old and New: How to compare domain independence** - WIPC 2019
Moraru, I. and Edelkamp, S.
Proc. of the *ICAPS 2019 Workshop on the International Planning Competition (WIPC)*.
- 2018 **Automated Pattern Selection using MiniZinc**
Martinez, M., **Moraru, I.** and Edelkamp, S.
Proc. of the *Int'l Conference on Principles and Practice of Constraint Programming: Workshop of Constraints and AI Planning*.
- 2023 **The ComplementaryPDB Planner**
Moraru, I., Edelkamp, S. and Franco, S.
Proc. of the *ICAPS 2023: IPC-10 Planner Abstracts* July, 2023
- 2018 **The Complementary2 Planner**
Franco, S., Lelis, L.H., Barley, M., Edelkamp, S., Martines, M. and **Moraru, I.**
Proc. of the *ICAPS 2018: IPC-9 Planner Abstracts*, pp.28-31.
- 2018 **The Planning-PDBs Planner**
Martinez, M., **Moraru, I.**, Edelkamp, S. and Franco, S.
Proc. of the *ICAPS 2018: IPC-9 Planner Abstracts*, pp.63-66.

TEACHING

- 2022 – 2024 **University of Lincoln, School of Computer Science - Lecturer**
INTRODUCTION TO DATA MINING for the Data Science and Applied Mathematics MSc.
ROBOTICS AND SUSTAINABILITY for the Robotics BSc.
- 2022 – 2024 **University of Lincoln, School of Computer Science - Assistant Lecturer**
DATA MINING for the Data Science MSc course.
- 2024 **University of Lincoln, Lincoln Institute for Agri-Foods Technology - Lecturer**
RESEARCH METHODS for several MSc courses.
- 2016 – 2021 **King's College London, Dept. of Informatics - Teaching Assistant**
MACHINE LEARNING AND PATTERN RECOGNITION (2018, 2019, 2020, 2021)
ARTIFICIAL INTELLIGENCE REASONING AND DECISION MAKING (2018, 2019, 2020)
ETHICS AND PHILOSOPHY OF ARTIFICIAL INTELLIGENCE (2019, 2020)
ARTIFICIAL INTELLIGENCE PLANNING (2018, 2019)
SOFTWARE ENGINEERING OF INTERNET APPLICATIONS (2017, 2018)
PROGRAMMING PRACTICE AND APPLICATIONS (2016, 2017)
SOFTWARE ENGINEERING GROUP PROJECT (2016)
PROGRAMMING PRACTICE (2016)
- 2020 – 2022 **King's College London Mathematics School**
Created and delivered the MACHINE LEARNING class for their Curriculum X term(2020/23)

TECHNICAL SKILLS

- OBJECT-ORIENTED PROGRAMMING Proficient: *Python, Java*
Intermediate: *C++, Ruby and Borland Pascal.*
Beginner: *C (MPI and CUDA)*
- WEB DEVELOPMENT Proficient: *JavaScript (NodeJS, React and React-Native), HTML and CSS*
Intermediate: *PHP and SQL*
Beginner: *MATLAB and R*
- MACHINE LEARNING Proficient: *Azure ML Services and SciKit-Learn*
Intermediate: *AWS, PyTorch and Keras*
- MODELING LANGUAGES Proficient: *PDDL*
Intermediate: *MiniZinc*
- ROBOTICS Proficient: *ROS1, ROSPlan and Bash*
Intermediate: *ROS2, MQTT, MavROS*
- LOGIC Intermediate: *Prolog*
- REVIEWER Knowledge Engineering Review (2019/23); International Conference on Automated Planning and Scheduling (2020); European Conference on Artificial Intelligence (2024-25); Explainable Robotics workshop at ICRA (2023)

AWARDS

- 2018 – 2021 **King’s Education Award - Nominated**
King’s College London, Department of Informatics
- 2018 **International Conference on Planning and Scheduling, Delft, The Netherlands**
International Planning Competition - 2nd Place in the Optimal Track
Planning and Execution Competition for Logistics Robots - 2nd Place Overall
- 2014 **UnituHack - 2nd Prize**
University Student and Staff Hackathon, London

SELECTED TALKS

- 2019 – 2024 **University of Trasilvania**
“AI Planning with Pattern Databases” – Invited Talk
Brasov, Romania
- 2018 **University of California, Berkeley, ICAPS 2019**
“Simplifying Automated Pattern Selection for Planning with Symbolic PDBs”
Berkely, US